23CSE373 · COMPUTER VISION · GROUP 10

Fast Object Proposal Generation Using Handcrafted Gradient Features

A from-scratch re-implementation and study of BING — Binarized Normed Gradients for Objectness Estimation (Cheng, Zhang, Lin, Torr · CVPR 2014)

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Why object proposals?

- **Object detection is expensive.** A sliding-window detector must classify every position × scale × aspect ratio, which is 10^5–10^6 windows per image.
- **Most windows are background.** Only a handful of windows actually contain an object.
- **Object proposals.** A cheap, class-agnostic objectness score ranks the windows, so the detector only examines the top ~1000.
- **Base paper: BING (CVPR 2014).** Handcrafted gradient features + bit operations produce proposals at 300 fps, with no deep learning.

$\sim 10^5+$

windows a naïve sliding-window scan evaluates per image

~1000

proposals that a detector needs to check instead

300 fps

BING's reported speed in the paper's C++ implementation

Problem definition

Given an image I, output a ranked list of N bounding boxes such that as many objects as possible are covered (IoU ≥ 0.5) by a small N, in milliseconds, using only handcrafted features.

1

Input

An RGB image of any size, any content

2

Output

Boxes (x_1, y_1, x_2, y_2) ranked by objectness score, plus an objectness heat-map

3

Constraints

Class-agnostic, real-time, CPU only, no deep networks

4

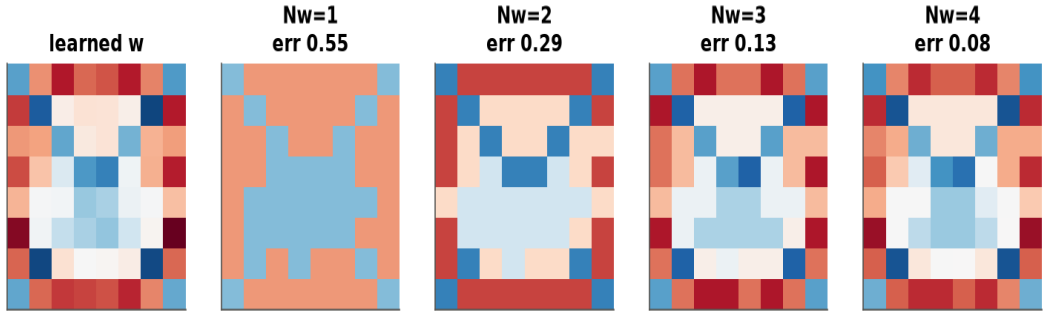
Success measures

Detection rate vs #proposals (DR@N), MABO (box tightness), runtime (fps)

Scope: we answer "Which regions are likely to contain objects?", not "What is the object?" (classification is a later stage).

Proposed solution: BING, rebuilt and extended

- **Key idea.** Objects have closed boundaries. Shrink any window to 8×8 and look at gradient magnitudes: object windows show a ring of edges.
- **64-weight linear model.** An SVM filter learns this “closed boundary” template (right); a second stage calibrates each window size.
- **Bits instead of floats.** Filter and features are binarized, so each window costs a few AND + POPCOUNT operations.



Our novelty beyond the paper

Stage III re-ranking

Colour, edge and geometry cues re-order BING's windows

Complete detector

HOG or CNN features + linear SVMs on our proposals, scored by VOC mAP

Head-to-head on VOC

Authors' C++ code, same images and evaluator; two missing details found

From-scratch Python

Numba kernels of the paper's Alg. 2, now multi-threaded

System design: 8-stage pipeline

1

Input image

longer side ≤ 500 px

2

Pre-processing

colour space: RGB / HSV / Gray / Lab

3

Candidates

36 power-of-two sizes (16…512, clipped), every 8×8 position

4

Gradient features

image resized per size, [-1 0 1] gradient

5

Binarized NG

top Ng=4 bit-planes packed into 64-bit words

6

Objectness score

s = ⟨w, g⟩ via AND + POPCOUNT (Nw=2)

7

Ranking

greedy NMS per score map, per-size calibration v·s + t

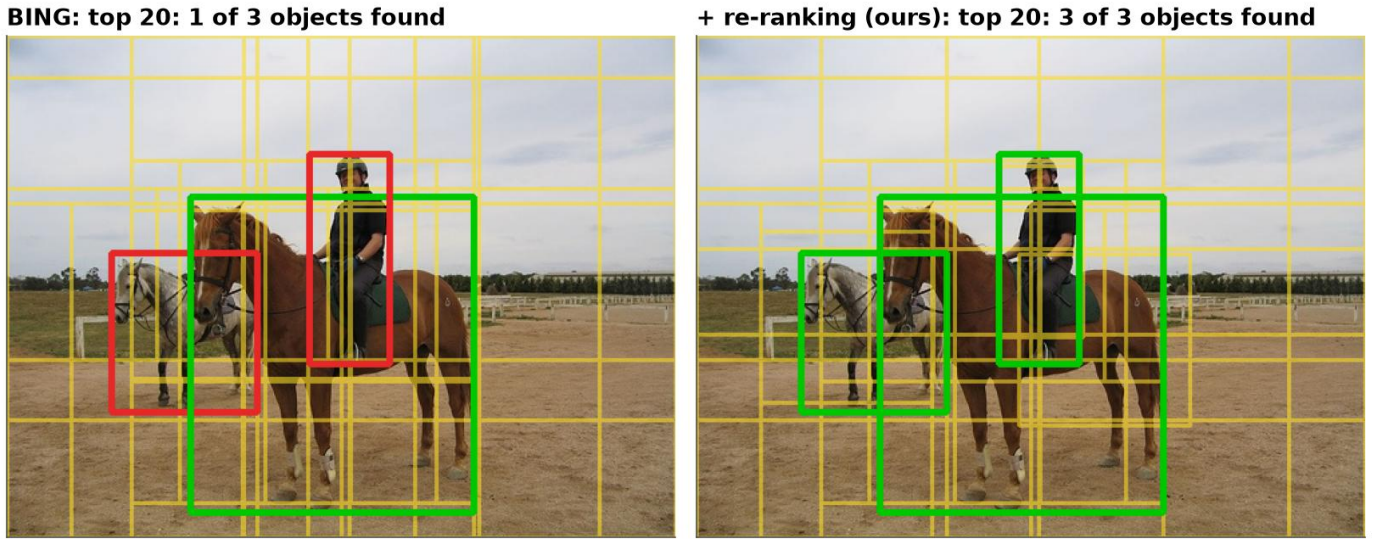
8

Top proposals

sorted boxes + objectness heat-map

Training: stage I is a linear SVM on 8×8 NG features of object boxes vs random background windows; stage II is a 1-D calibration per window size. Our stage III then re-ranks the output (next slide).

Our novelty: multi-cue re-ranking (stage III)



Top 20 windows (yellow) on a VOC 2007 test image. Green: object found (IoU ≥ 0.5); red: missed.

- **Problem.** BING ranks windows with a single cue, its 8×8 edge template, so object windows can sit far down the list.
- **Our idea.** Re-score BING's windows with extra cheap, handcrafted cues, learned from VOC trainval.

1 · Colour contrast
Lab colour inside the window vs. a ring around it

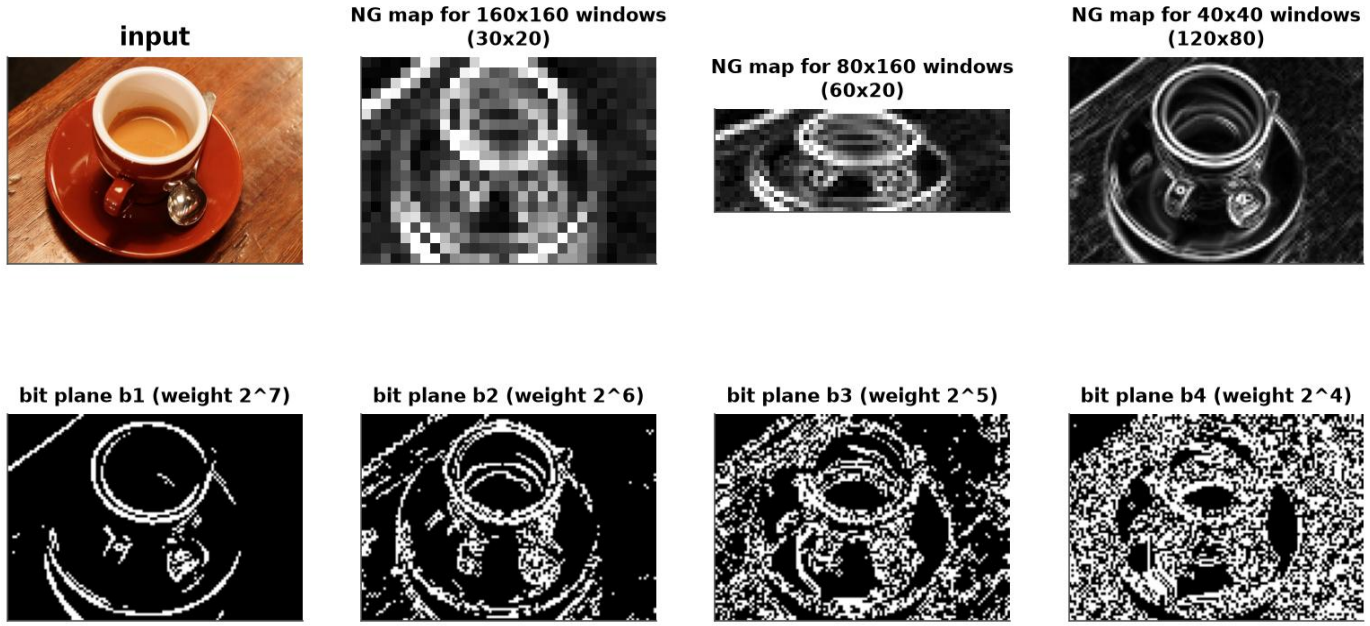
2 · Edge density
Gradient energy inside vs. the ring

3 · Geometry
Position, size and aspect ratio of the window

4 · BING itself
Its calibrated score and rank

A small gradient-boosted tree model combines the 16 values into a new score. Every cue is read from integral images in O(1): no deep network.

Normed gradients → bit operations



Top: NG maps for three window sizes (each 8×8 patch = one window). Bottom: the 4 bit-planes forming the binarized feature.

- Implemented in Python: bing/ package (features, binary, Numba fast path, model, metrics, datasets); unit tests prove bitwise score = float score.

Approximate the filter

$w \approx \Sigma_j \beta_j a_j, \quad a_j \in \{-1, +1\}^{64}$

Approximate the feature

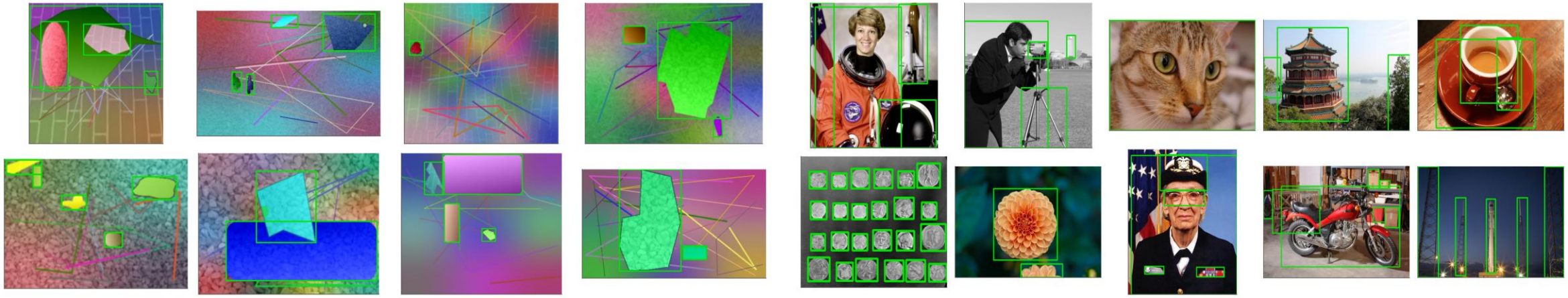
$g \approx \Sigma_k 2^{(8-k)} b_k, \quad b_k \in \{0, 1\}^{64}$

Score with bits only

$\langle a_j, b_k \rangle = 2 \cdot \text{popcnt}(a_j^+ \text{ AND } b_k) - \text{popcnt}(b_k)$

12 POPCOUNTs per window instead of 64 multiply-adds.

Experimental setup



PASCAL VOC 2007

The paper's benchmark: train on 5011 trainval images, test on 4952 test images (12,032 objects, “difficult” ones ignored).

Synthetic scenes

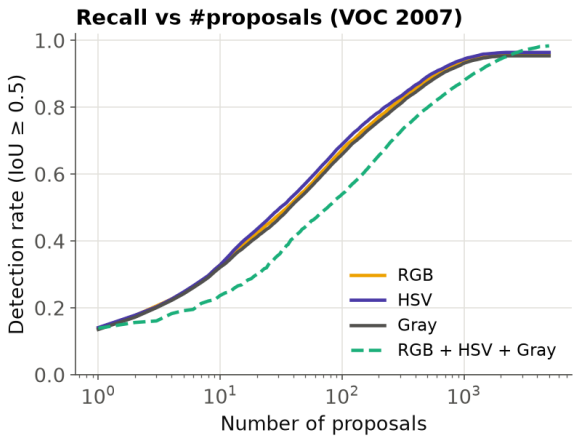
600 train / 300 test images: shapes with texture and shading on cluttered backgrounds (edges that are not objects).

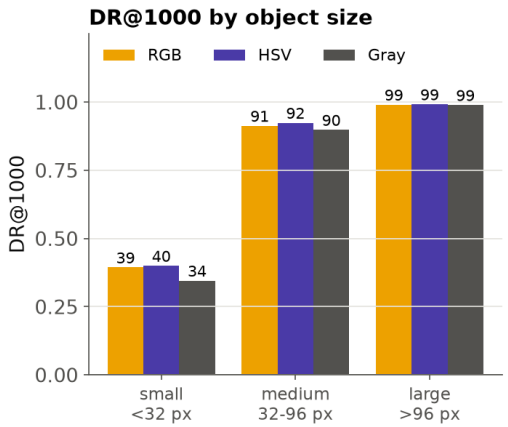
Real photographs

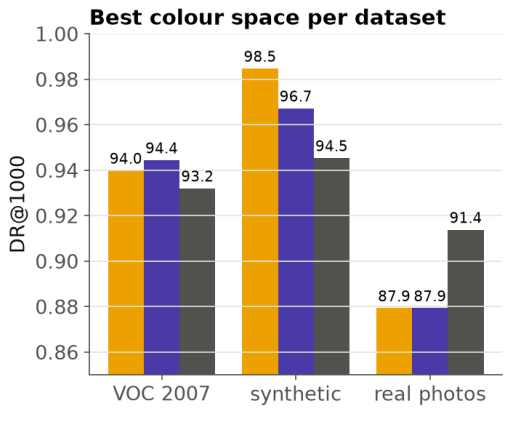
10 photos, 58 annotated objects, never used for training: tests generalisation.

Metrics: detection rate DR@N (share of objects covered with IoU ≥ 0.5 by the top N proposals), MABO (mean best overlap), images/s; for the detector, VOC mean average precision (mAP). Baselines: the authors' own C++ BING, Selective Search, sliding windows, random boxes. All scored with the same evaluator.

Design choice: RGB, HSV or Gray?







- **Setup:** one BING model per colour space, trained on VOC trainval; the gradient is the strongest of the channels.
- **VOC:** HSV is best and wins 15 of 20 classes at 100 boxes. Gray misses colour-only edges, most on small objects (34.5% vs 39.3%).
- **Other data:** RGB is best on the synthetic scenes; Gray transfers best from synthetic training to real photos.
- **Merging all three** (the paper's BING-Diversified) only pays off at large budgets.

94.4%

HSV: VOC objects in the top 1000 (RGB 94.0%, Gray 93.2%)

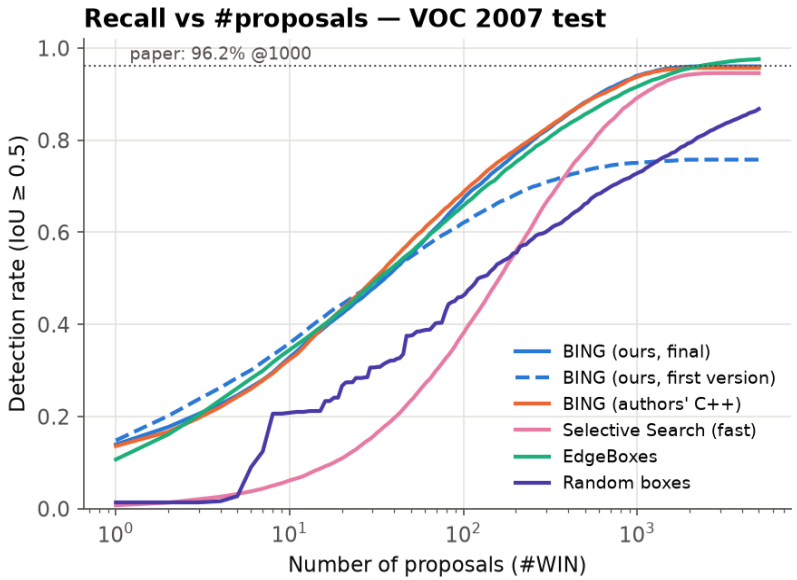
98.4%

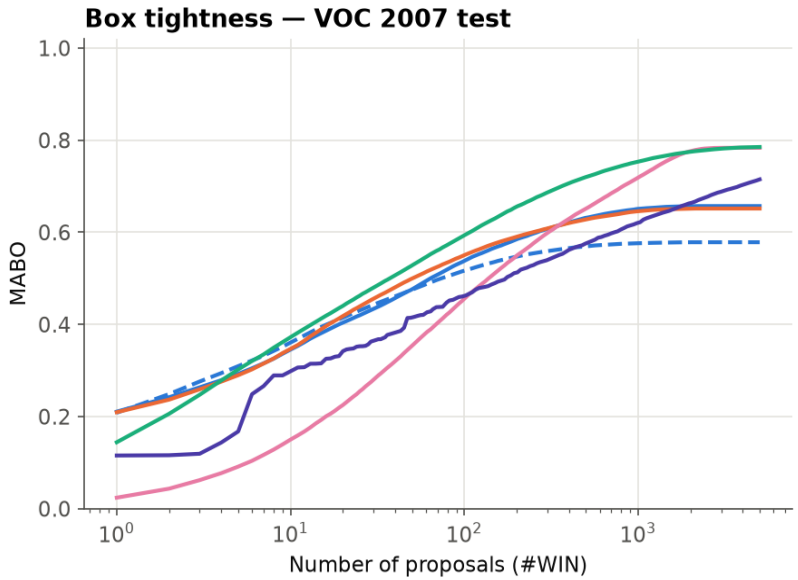
all three merged, top 5000 (best single: 96.4%)

91.4%

Gray on real photos, trained on synthetic only (RGB, HSV: 87.9%)

PASCAL VOC 2007: reproducing the paper





- **Our rebuild matches the authors' code** on their own benchmark: 94.0% vs 93.8% at 1000, 96.0% vs 95.7% at 5000; box tightness 0.651 vs 0.646.
- **By the authors' own evaluator** their code scores 96.0% here, so the paper's number reproduces.
- **Unseen classes (the paper's test):** trained on only 6 classes, BING finds 91.1% of objects of the other 14 (trained on all 20: 93.3%).

94.0%

objects found in the top 1000, our final BING

93.8%

the authors' C++ code, same images and evaluator

96.2%

reported in the paper (authors' evaluator)

Comparison with existing methods

Method	VOC DR@1k	VOC MABO	Synth. DR@1k	Real DR@1k	img/s (1 thread)
BING — ours (Python + Numba)	**94.0%**	**0.651**	**98.5%**	**87.9%**	**98**
BING — authors' C++ code	93.8%	0.646	94.9%	75.9%	76
Selective Search (fast)	89.2%	0.719	97.5%	100.0%	1.3
EdgeBoxes	91.6%	0.753	—	—	6.5
Random boxes	72.9%	0.621	53.6%	48.3%	—

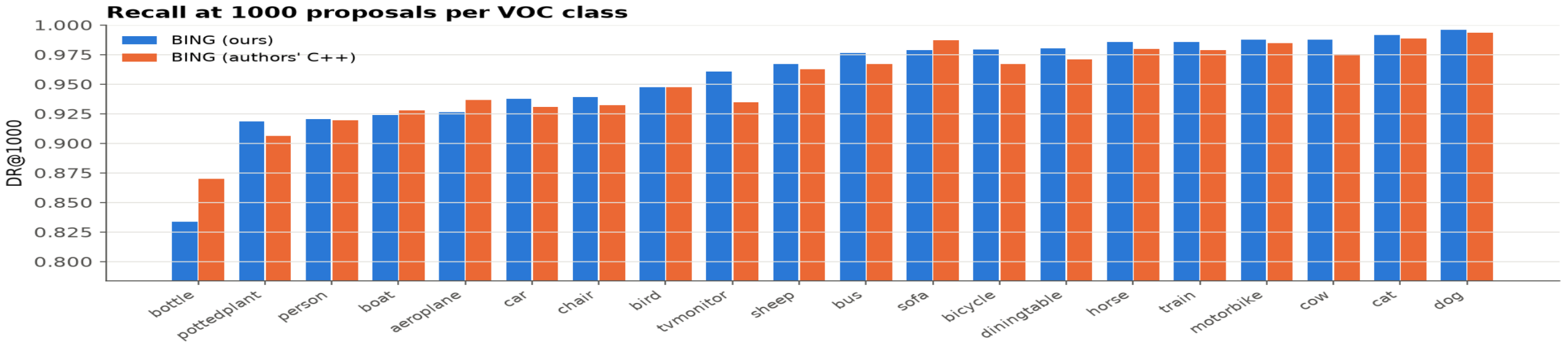
All methods on all 4,952 VOC test images; speed on one thread, all timed on the same machine. With 4 threads, ours reaches 294 img/s and the authors' code 300.

- **BING vs Selective Search and EdgeBoxes:** BING finds the most objects in the top 1000 (94.0% vs 89.2% and 91.6%) and is 15–77× faster, but its boxes are looser (MABO 0.65 vs 0.72 and 0.75).
- **Ours vs the authors' code:** same accuracy on VOC; ours generalises better to real photos when trained on synthetic data, and is faster on one thread (98 vs 76 img/s).

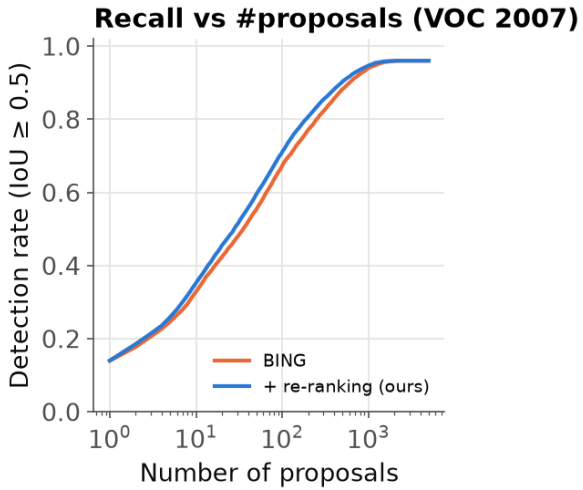
Two details the paper leaves out

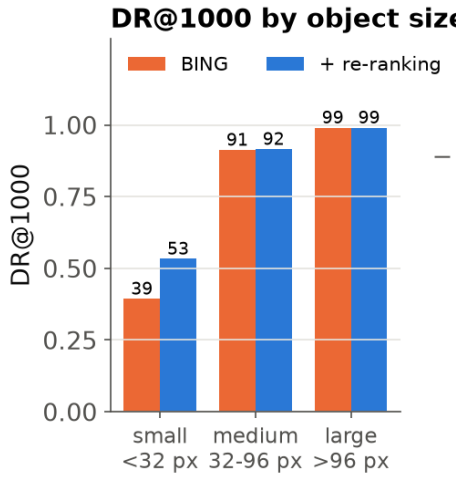
Our BING on VOC 2007	DR@1k	DR@5k	MABO
Paper's text: sizes 10–320, local-max NMS (first version)	75.1%	75.8%	0.576
+ released code's window sizes (16–512, clipped)	79.2%	79.7%	0.607
+ released code's greedy NMS only	92.2%	94.4%	0.621
Both (final)	**94.0%**	**96.0%**	**0.651**

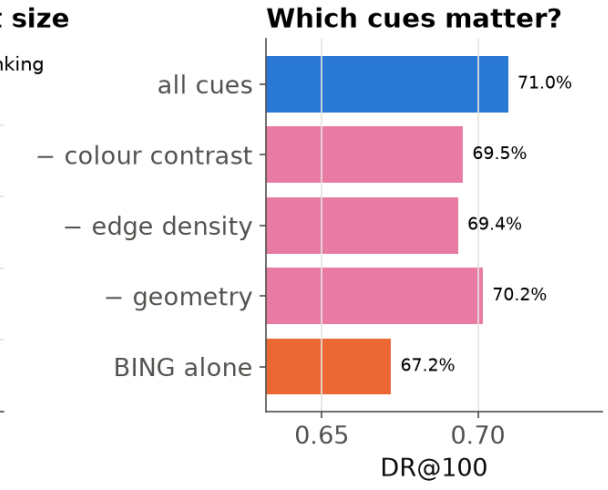
- **Window sizes:** the paper lists 10…320; the released code uses 16…512 and clips windows to the image, so near full-image boxes exist. VOC is full of large objects.
- **Suppression:** keeping only local maxima of each score map throws away good windows; the code's greedy suppression keeps the best 130 spread-out windows per size.
- **Lesson:** reproducing a paper needs its code, not just its text.



Novelty result: more objects, found sooner







- **Edge density is the key cue for small objects:** without it, small-object recall falls from 53.3% to 45.0%.
- **Cost:** 17 instead of 10 ms per image on one thread, still 46× faster than Selective Search.
- **Tried and rejected:** snapping box edges to strong gradients and adding finer window sizes; neither raised recall on the full test set.

71.0%

objects found in the top 100 (BING alone: 67.2%)

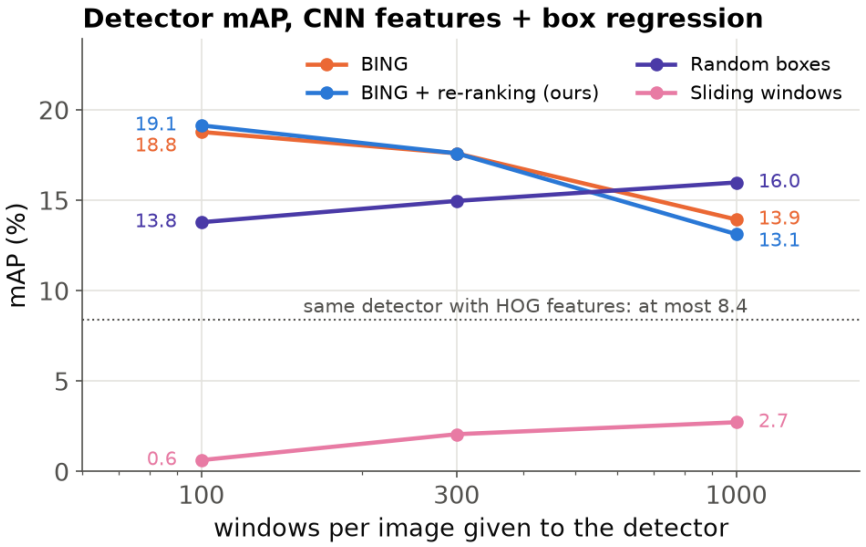
53.3%

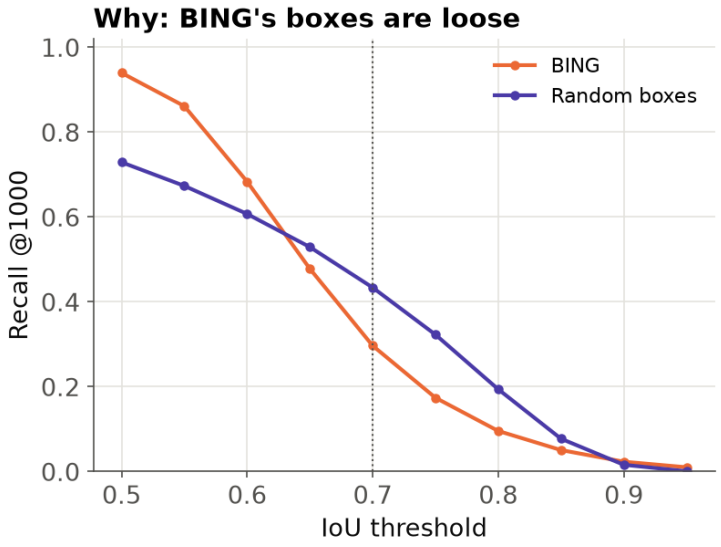
small objects found in the top 1000 (BING alone: 39.3%)

94.7%

objects found in the top 1000 (BING alone: 94.0%)

A complete detector built on our proposals





- **Pipeline:** each window → mean ResNet-50 conv4 + conv5 activations (ImageNet, frozen) → 20 linear SVMs + box regression. A fully handcrafted version (HOG + colour) reaches 7.4%.
- **Finding:** at 100 windows BING beats random boxes (18.8% vs 13.8%); at 1000, random boxes win (16.0%) because BING's boxes are loose. Hosang et al. (TPAMI 2016): recall at stricter overlaps predicts detection quality.

19.1%

mAP, CNN detector on our re-ranked top 100 (HOG: 7.4%)

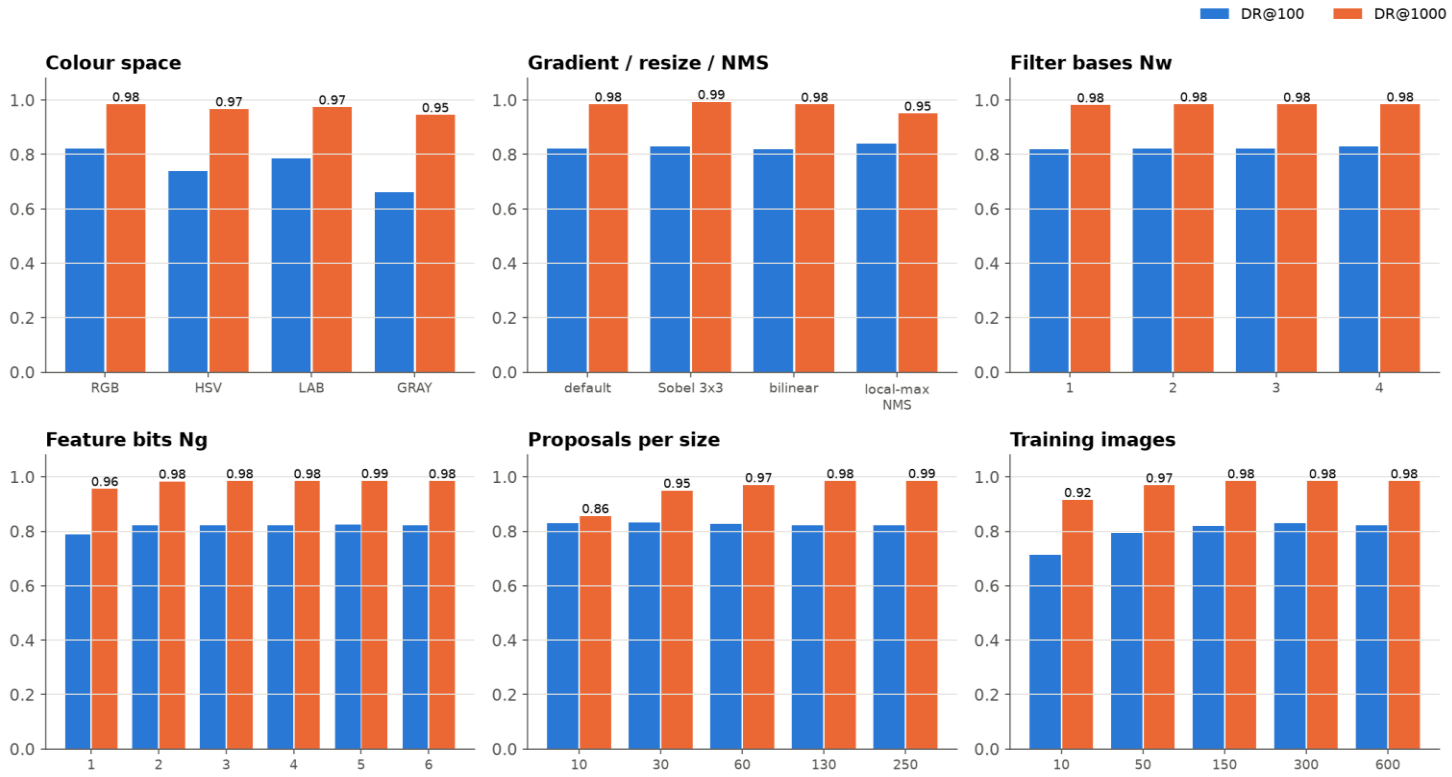
0.6%

mAP from 100 sliding windows, same detector (30× lower)

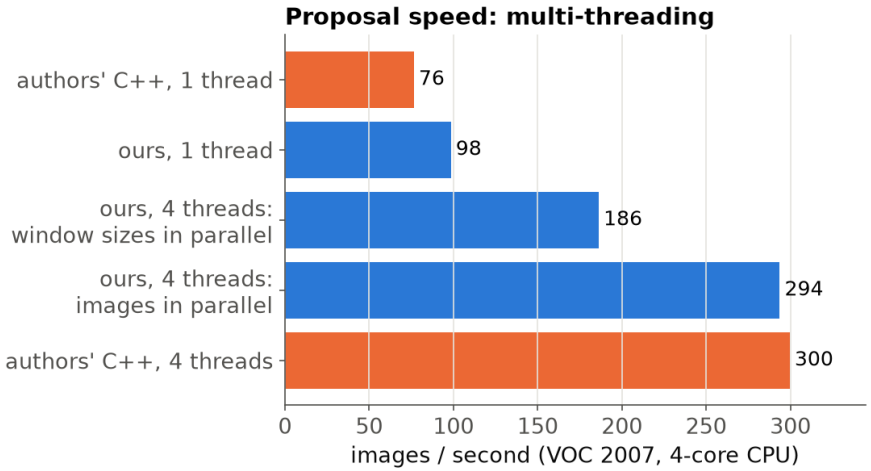
30%

of objects covered at IoU ≥ 0.7 by BING's top 1000 (random boxes: 43%)

What matters? Ablations and speed



DR@1000 on the synthetic test set (bars: blue DR@100, orange DR@1000).



- **Gradient:** [-1 0 1] 98.5% ≈ Sobel 99.3%
- **Colour:** RGB 98.5% > Lab 97.4% > HSV 96.7% > Gray 94.5%
- **Bits:** Ng = 1 → 95.6%; Ng ≥ 2 → 98.4%
- **Data:** 10 images → 91.6%; 600 → 98.5%
- **Speed:** 4 threads, one image each → 294 img/s (authors' C++ on the same machine: 300)

Limitation: power-of-two sizes and an 8-px stride give coarse boxes (VOC MABO 0.65); small objects such as bottles are hardest.

CONCLUSION

Gradients, a 64-weight model and bit operations find 94% of VOC objects, as the authors' code does; our re-ranking finds them sooner.

94.0%

VOC 2007 objects found in top 1000 (authors' code 93.8%)

71.0%

found in the top 100 with our re-ranking (BING: 67.2%)

53.3%

small objects in the top 1000 with re-ranking (BING: 39.3%)

294

images / second, 4 threads (authors' C++: 300)

Future work tighter boxes (better box regression, finer sizes) · fine-tuned CNN features · train-on-some-classes generalisation

References [1] M.-M. Cheng, Z. Zhang, W.-Y. Lin, P. H. S. Torr, "BING: Binarized Normed Gradients for Objectness Estimation at 300fps," CVPR 2014. [2] J. Uijlings et al., "Selective Search for Object Recognition," IJCV 2013. [3] B. Alexe, T. Deselaers, V. Ferrari, "Measuring the Objectness of Image Windows," TPAMI 2012. [4] J. Hosang et al., "What makes for effective detection proposals?," TPAMI 2016.

Thank you